Results

Results I

Results are interpreted only after the registered analysis has run and the deviation ledger (Section 6) has classified any departures from the plan. Every value below is read from `output/data/demo_analysis.json`, which is regenerated by `scripts/generate_figures.py` from the tested `run_registered_analysis` function; the sensitivity row in the final subsection is read from the frozen registration's `sensitivity_analyses`, the same source `build_review_packet` consumes.

Confirmatory outcome (H1, `primary_score`) I

On the deterministic demonstration data, the registered two-sided permutation test returns:

Confirmatory outcome (H1, `primary_score`) I

Quantity	Value
Control mean	-0.178991
Treatment mean	0.823948
Observed mean difference	1.002940
Permutations	2000
Shuffles with absolute difference $\geq$ observed	0
Two-sided permutation p-value	0.0005
Preregistered `alpha`	0.05
Confirmatory decision	significant

Confirmatory outcome (H1, primary_score) I

The observed difference of 1.003 lies far in the right tail of the permutation null distribution: none of the 2000 label shuffles produced a mean difference as extreme, so the add-one-corrected p-value is 0.0005, below the preregistered alpha = 0.05. H1 is therefore **supported** as a confirmatory result. fig. 1 shows the observed statistic against the null it was tested against.

Confirmatory outcome (H1, `primary_score`) I

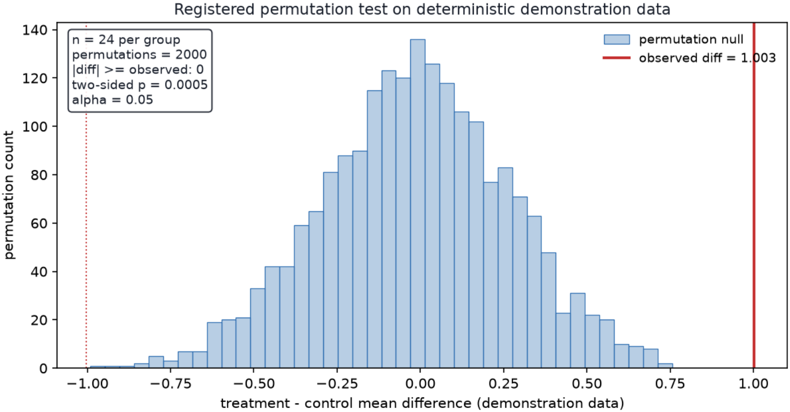

Figure 1: Registered permutation test on the deterministic demonstration data, rendered by `plot_permutation_result` from `permutation_null_distribution` and `run_permutation_test`. The histogram is the label-permutation null distribution of the treatment-minus-control mean difference (2000 seeded shuffles); the solid red line marks the observed difference of `1.003` and the dotted line its mirror image. The annotation reports `n = 24` per group, `0` shuffles at least as extreme, a two-sided p-value of `0.0005`, and `alpha = 0.05`. The observed statistic sits outside the entire null, which is why the primary confirmatory claim holds.

Claim boundary I

The registration preregisters exactly one confirmatory outcome. The review packet built by `build_review_packet` records `primary_score` as the sole confirmatory outcome and returns an integrity score of `1.0` for a plan-faithful execution. Any other endpoint examined during analysis is exploratory and is reported under Section 6, never here. This separation is enforced in code, not merely asserted in prose.

Sensitivity analysis I

The frozen registration declares one preregistered sensitivity analysis, and `validate_sensitivity_table` checks every declared row against the registered outcomes and primary model before the review packet is built:

Sensitivity analysis I

Analysis	Target	Model	Decision
label_permutation_sensitivity	primary_score	permutation_test	robust

Sensitivity analysis I

The row targets the registered primary outcome with the registered primary model and a supported decision, so `validate_sensitivity_table` returns no findings for it and `output/reports/sensitivity_findings.json` records an empty findings list. The same row is carried verbatim into the review packet (`build_review_packet` reads `sensitivity_analyses` from the frozen plan), so the sensitivity check is visible in source, tests, review packets, and this manuscript — never invented after results are known.